

BPS-17 Lecturer Mathematics

AJKPSC / FPSC / PPSC / KPSC

Comprehensive Study Plan

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Abstract Algebra & Linear Algebra

Group Theory • Ring Theory • Vector Spaces • Linear Algebra

Topics Covered:

- Groups: Definition & Examples
- Abelian & Non-Abelian Groups
- Lagrange's Theorem
- Normalizer & Centralizer
- Automorphisms
- Rings, Subrings, Integral Domains
- Vector Spaces
- Basis & Dimension
- Finite Fields
- Order of a Group & Element Order
- Cyclic Groups & Generators
- Normal & Characteristic Subgroups
- Homomorphism & Isomorphism
- Isomorphism Theorems
- Fields & Field Extensions
- Linear Independence & Dependence
- Finite & Infinite Dimensional Spaces
- Linear Transformations

Each section: **Concept Summary** → **Key Formulas** → **Past MCQs**
with Full Explanations

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1. Groups: Definition, Examples & Basic Properties

A non-empty set G with a binary operation $*$ is a **group** $(G, *)$ if:

G1. Closure: $\forall a, b \in G, a * b \in G$

G2. Associativity: $\forall a, b, c \in G, (a * b) * c = a * (b * c)$

G3. Identity: $\exists e \in G$ such that $a * e = e * a = a \quad \forall a \in G$

G4. Inverse: $\forall a \in G, \exists a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

If additionally $a * b = b * a \quad \forall a, b \in G$, then G is **Abelian (commutative)**.

Group Examples:

- $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$
- $(\mathbb{Q}^*, \cdot)$, $(\mathbb{R}^*, \cdot)$, $(\mathbb{C}^*, \cdot)$
- $(\mathbb{Z}_n, +_n)$ — integers mod n
- $\text{GL}(n, \mathbb{R})$ — invertible $n \times n$ matrices
- S_n — symmetric group (permutations)
- D_n — dihedral group (symmetries)

Non-Group Examples:

- $(\mathbb{N}, +)$ — no inverse
- $(\mathbb{Z}, \cdot)$ — no inverse for $n \neq \pm 1$
- $(\mathbb{Z}_n, \cdot)$ — not closed under inverse in general
- $(M_n(\mathbb{R}), \cdot)$ — no inverse for singular matrices

Key Formulas & Results

Key Facts: Identity is **unique**. Inverses are **unique**. $(ab)^{-1} = b^{-1}a^{-1}$.

Cancellation Laws: $ab = ac \Rightarrow b = c$ (left); $ba = ca \Rightarrow b = c$ (right).

Order of element a : smallest positive n with $a^n = e$. Written $\text{ord}(a) = n$.

1.1. MCQs — Groups: Definition & Examples

Q1. Which of the following is **NOT** a group under the given operation?

- (A) $(\mathbb{Z}, +)$
- (B) $(\mathbb{Z}, \times)$
- (C) $(\mathbb{Q}^*, \times)$
- (D) $(\mathbb{R}, +)$

Answer: (B) $(\mathbb{Z}, \times)$

In $(\mathbb{Z}, \times)$, the integer 2 has no multiplicative inverse in $\mathbb{Z}$ (since $1/2 \notin \mathbb{Z}$). All

other options satisfy all four group axioms. $(\mathbb{Z}, +)$, $(\mathbb{Q}^*, \times)$, and $(\mathbb{R}, +)$ are classical groups.

Q2. The set of all 2×2 invertible matrices over $\mathbb{R}$ under matrix multiplication forms:

- (A) An Abelian group
- (B) A non-Abelian group
- (C) A ring but not a group
- (D) Not a group

Answer: (B) A non-Abelian group

$GL(2, \mathbb{R})$ (General Linear Group) satisfies all group axioms. It is **non-Abelian** because matrix multiplication is not commutative in general: $AB \neq BA$ for many invertible matrices A, B .

Q3. In a group G , the inverse of the identity element e is:

- (A) $-e$
- (B) e^{-1}
- (C) e
- (D) 0

Answer: (C) e

Since $e \cdot e = e$, the element e is its own inverse: $e^{-1} = e$. This follows directly from the definition of identity and inverse.

Q4. Which of the following groups is **Abelian**?

- (A) S_3 (symmetric group on 3 letters)
- (B) D_4 (dihedral group of square)
- (C) $(\mathbb{Z}_6, +)$
- (D) $GL(2, \mathbb{R})$

Answer: (C) $(\mathbb{Z}_6, +)$

$(\mathbb{Z}_6, +)$ is cyclic and therefore Abelian: $a + b = b + a$ for all $a, b \in \mathbb{Z}_6$. Options (A), (B), (D) are all non-Abelian: S_3 has order 6 and is the smallest non-Abelian group; D_4 has non-commuting rotations and reflections.

Q5. The order of the element 2 in $(\mathbb{Z}_8, +)$ is:

- (A) 2
- (B) 4

- (C) 8
(D) 1

Answer: (B) 4

We need the smallest positive n with $n \cdot 2 \equiv 0 \pmod{8}$:

$1 \cdot 2 = 2$, $2 \cdot 2 = 4$, $3 \cdot 2 = 6$, $4 \cdot 2 = 8 \equiv 0$. So $\text{ord}(2) = 4$.

Formula: In $\mathbb{Z}_n$, $\text{ord}(k) = \frac{n}{\gcd(n, k)} = \frac{8}{\gcd(8, 2)} = \frac{8}{2} = 4$.

Order formula in $\mathbb{Z}_n$: $\text{ord}(k) = \frac{n}{\gcd(n, k)}$. This is one of the most frequently tested results in PSC exams on group theory. Memorize it!

Q6. The order of the symmetric group S_n is:

- (A) n
(B) $2n$
(C) $n!$
(D) n^2

Answer: (C) $n!$

S_n consists of all permutations of n objects. The number of such permutations is $n!$. For example: $|S_3| = 6$, $|S_4| = 24$, $|S_5| = 120$.

Q7. If a is an element of a group G and $\text{ord}(a) = n$, then $\text{ord}(a^k)$ equals:

- (A) nk
(B) $\frac{n}{\gcd(n, k)}$
(C) $\gcd(n, k)$
(D) $\frac{k}{n}$

Answer: (B) $\frac{n}{\gcd(n, k)}$

This is a standard theorem: if $\text{ord}(a) = n$ in a group, then $\text{ord}(a^k) = \frac{n}{\gcd(n, k)}$.

Example: In $\mathbb{Z}_{12}$, $\text{ord}(1) = 12$. Then $\text{ord}(1^4) = \text{ord}(4) = \frac{12}{\gcd(12, 4)} = \frac{12}{4} = 3$.

Check: $4 + 4 + 4 = 12 \equiv 0$.

2. Cyclic Groups

A group G is **cyclic** if it is generated by a single element: $G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$.

- Every cyclic group is **Abelian**.
- A finite cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +)$.
- An infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$.
- The number of generators of $\mathbb{Z}_n$ is $\phi(n)$ (Euler's totient function).
- Every **subgroup** of a cyclic group is cyclic.
- For finite cyclic $G = \langle a \rangle$ of order n : $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$ and has order $\frac{n}{\gcd(n,k)}$.

Key Formulas & Results

Euler's Totient: $\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$ where the product is over distinct prime factors.

Generators of $\mathbb{Z}_n$: k generates $\mathbb{Z}_n \Leftrightarrow \gcd(k, n) = 1$.

Subgroups of $\mathbb{Z}_n$: One subgroup of order d for each $d \mid n$.

2.1. MCQs — Cyclic Groups

Q8. Which of the following is **not** a cyclic group?

- (A) $(\mathbb{Z}, +)$
- (B) $(\mathbb{Z}_6, +)$
- (C) S_3
- (D) $(\mathbb{Z}_p, +)$ for prime p

Answer: (C) S_3

S_3 has order 6 and is non-Abelian. Since every cyclic group is Abelian, S_3 cannot be cyclic. All other options are cyclic: $\mathbb{Z} = \langle 1 \rangle$, $\mathbb{Z}_6 = \langle 1 \rangle$, and $\mathbb{Z}_p = \langle 1 \rangle$ for prime p .

Q9. The number of generators of the cyclic group $\mathbb{Z}_{12}$ is:

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Answer: (C) 4

Number of generators $= \phi(12) = 12 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4$.

The generators are elements k with $\gcd(k, 12) = 1$: namely 1, 5, 7, 11.

Q10. Every subgroup of a cyclic group is:

- (A) Normal
- (B) Abelian
- (C) Cyclic
- (D) All of the above

Answer: (D) All of the above

Every subgroup of a cyclic group is itself cyclic, hence Abelian. In an Abelian group, every subgroup is normal (since $gHg^{-1} = H$ trivially when G is Abelian). So all three properties hold.

Q11. The cyclic group $\mathbb{Z}_p$ for prime p has exactly how many subgroups?

- (A) p
- (B) 2
- (C) $p - 1$
- (D) 1

Answer: (B) 2

By Lagrange's theorem, subgroup orders must divide $|G| = p$. Since p is prime, its only positive divisors are 1 and p . So $\mathbb{Z}_p$ has exactly two subgroups: the trivial group $\{0\}$ and $\mathbb{Z}_p$ itself.

Q12. If $G = \langle a \rangle$ is cyclic of order 12, then the order of the subgroup $\langle a^8 \rangle$ is:

- (A) 8
- (B) 4
- (C) 3
- (D) 2

Answer: (C) 3

$\text{ord}(a^8) = \frac{12}{\gcd(12, 8)} = \frac{12}{4} = 3$. So $\langle a^8 \rangle$ has order 3.

Check: $(a^8)^3 = a^{24} = (a^{12})^2 = e^2 = e \checkmark$, and $(a^8)^1, (a^8)^2 \neq e$.

3. Lagrange's Theorem & Subgroups

If H is a subgroup of a finite group G , then $|H|$ divides $|G|$.

Moreover: **Index of H in G** $= [G : H] = \frac{|G|}{|H|}$ (number of distinct cosets of H).

- $\text{ord}(a)$ divides $|G|$ for all $a \in G$ (element order divides group order).
- If $|G| = p$ (prime), then G is cyclic and has no proper non-trivial subgroups.
- $a^{|G|} = e$ for all $a \in G$ (**Fermat–Euler** consequence).
- **Converse is FALSE:** $|H|$ divides $|G|$ does not guarantee a subgroup of order $|H|$ exists (example: A_4 has order 12 but no subgroup of order 6).

3.1. MCQs — Lagrange's Theorem

Q13. A group of prime order p must be:

- (A) Non-Abelian
- (B) Cyclic and Abelian
- (C) Have a proper non-trivial subgroup
- (D) Simple and non-cyclic

Answer: (B) Cyclic and Abelian

If $|G| = p$ (prime), Lagrange's theorem forces all element orders to be 1 or p . Any element $a \neq e$ has order p , so $G = \langle a \rangle$ is cyclic. All cyclic groups are Abelian.

Q14. If $|G| = 35$, which of the following could **not** be the order of a subgroup of G ?

- (A) 5
- (B) 7
- (C) 35
- (D) 15

Answer: (D) 15

By Lagrange's theorem, subgroup orders must divide $|G| = 35 = 5 \times 7$. The divisors of 35 are: 1, 5, 7, 35. Since $15 \nmid 35$, no subgroup of order 15 can exist.

Q15. In a group G of order 24, the possible orders of elements are:

- (A) Only 1, 2, 3, 4

- (B) Divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24
 (C) Any positive integer
 (D) Only prime divisors of 24

Answer: (B) Divisors of 24: 1, 2, 3, 4, 6, 8, 12, 24

By Lagrange's theorem, $\text{ord}(a)$ must divide $|G| = 24$. The positive divisors of 24 are 1, 2, 3, 4, 6, 8, 12, 24, so these are the only possible element orders.

Q16. The index $[S_4 : A_4]$ (where A_4 is the alternating group) equals:

- (A) 3
 (B) 4
 (C) 2
 (D) 12

Answer: (C) 2

$|S_4| = 24$, $|A_4| = 12$. Index $= \frac{|S_4|}{|A_4|} = \frac{24}{12} = 2$. Any subgroup of index 2 in any group is automatically normal—a key fact used repeatedly.

Key PSC Trick: A subgroup H of G has index 2 $\Rightarrow H$ is **normal** in G . This is because there are exactly two cosets (H and $G \setminus H$), so both left and right cosets must coincide.

4. Normal Subgroups, Normalizer & Centralizer

A subgroup N of G is **normal** ($N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$, equivalently $gN = Ng$ (left and right cosets coincide).

- Every subgroup of an Abelian group is normal.
- $\{e\}$ and G itself are always normal in G .
- Index-2 subgroups are always normal.
- $A_n \trianglelefteq S_n$ for all n .

Normalizer & Centralizer

For a subset $H \subseteq G$:

- **Normalizer:** $N_G(H) = \{g \in G : gHg^{-1} = H\}$ — the largest subgroup of G

in which H is normal.

- **Centralizer of H :** $C_G(H) = \{g \in G : gh = hg \ \forall h \in H\}$ — elements of G commuting with every element of H .
- **Centre of G :** $Z(G) = C_G(G) = \{g \in G : gx = xg \ \forall x \in G\}$.
- $C_G(H) \subseteq N_G(H) \subseteq G$.
- G is Abelian $\Leftrightarrow Z(G) = G$.

4.1. MCQs — Normal Subgroups, Normalizer, Centralizer

Q17. A subgroup N of G is normal if and only if:

- (A) N is Abelian
- (B) $gNg^{-1} = N$ for all $g \in G$
- (C) N has prime order
- (D) N has index 3

Answer: (B) $gNg^{-1} = N$ for all $g \in G$

This is the definition of a normal subgroup. Note: option (D) would be true only for index 2, not 3 in general. Normality has nothing to do with N being Abelian or having prime order.

Q18. The centre $Z(G)$ of a group G is always:

- (A) Trivial
- (B) Normal in G
- (C) Non-Abelian
- (D) Equal to G

Answer: (B) Normal in G

$Z(G)$ is always a normal subgroup of G . Proof: for $z \in Z(G)$ and $g \in G$: $gzg^{-1} = gg^{-1}z = z \in Z(G)$. $Z(G) = G$ iff G is Abelian (not always). $Z(G)$ need not be trivial.

Q19. If G is Abelian, then for any $a \in G$, the centralizer $C_G(a)$ is:

- (A) $\{e\}$
- (B) $\langle a \rangle$
- (C) G
- (D) $Z(G)$ only

Answer: (C) G

In an Abelian group, $gag^{-1} = a$ for all $g \in G$, so every element commutes with a . Thus $C_G(a) = G$. Equivalently, $Z(G) = G$ for Abelian groups.

Q20. Which of the following is a normal subgroup of S_3 ?

- (A) $\langle(12)\rangle$
- (B) $\langle(13)\rangle$
- (C) $\langle(123)\rangle$
- (D) $\langle(23)\rangle$

Answer: (C) $\langle(123)\rangle$

$\langle(123)\rangle = \{e, (123), (132)\} = A_3$ is the alternating group $A_3 \trianglelefteq S_3$. It has index 2 in S_3 (since $|S_3| = 6$, $|A_3| = 3$), so it is normal. The transposition-generated subgroups $\langle(12)\rangle$ etc. have order 2 and index 3; they are **not** normal in S_3 .

Q21. The normalizer $N_G(H)$ of a subgroup H in G is:

- (A) Always equal to H
- (B) The smallest subgroup containing H
- (C) The largest subgroup of G in which H is normal
- (D) Equal to the centralizer $C_G(H)$

Answer: (C) The largest subgroup of G in which H is normal

By definition, $N_G(H) = \{g \in G : gHg^{-1} = H\}$. This is the largest subgroup of G such that $H \trianglelefteq N_G(H)$. In general $C_G(H) \subseteq N_G(H)$ but they need not be equal.

5. Homomorphism, Isomorphism & Automorphisms

A map $\phi : G \rightarrow H$ is a **group homomorphism** if $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$.

- $\phi(e_G) = e_H$ and $\phi(a^{-1}) = (\phi(a))^{-1}$.
- **Kernel:** $\text{Ker}(\phi) = \{g \in G : \phi(g) = e_H\}$ — always a **normal** subgroup of G .
- **Image:** $\text{Im}(\phi) = \{\phi(g) : g \in G\}$ — a subgroup of H .
- ϕ is **injective** $\Leftrightarrow \text{Ker}(\phi) = \{e_G\}$.
- ϕ is an **isomorphism** if bijective homomorphism. Write $G \cong H$.
- ϕ is an **automorphism** if it is an isomorphism from G to itself.

If $\phi : G \rightarrow H$ is a group homomorphism, then:

$$G / \text{Ker}(\phi) \cong \text{Im}(\phi)$$

In particular: if ϕ is surjective, then $G / \text{Ker}(\phi) \cong H$.

Second (Diamond): If $H \leq G$ and $N \trianglelefteq G$, then $HN \leq G$, $H \cap N \trianglelefteq H$, and

$$\frac{HN}{N} \cong \frac{H}{H \cap N}$$

Third (Quotient of Quotients): If $N \trianglelefteq K \trianglelefteq G$ and $N \trianglelefteq G$, then

$$\frac{G/N}{K/N} \cong \frac{G}{K}$$

5.1. MCQs — Homomorphism & Isomorphism

Q22. If $\phi : G \rightarrow H$ is a group homomorphism, then $\text{Ker}(\phi)$ is:

- (A) A normal subgroup of H
- (B) A normal subgroup of G
- (C) The image of G in H
- (D) A subgroup of H

Answer: (B) A normal subgroup of G

The kernel is always normal in the *domain*: for $k \in \text{Ker}(\phi)$ and $g \in G$, $\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g)^{-1} = \phi(g)e_H\phi(g)^{-1} = e_H$, so $gkg^{-1} \in \text{Ker}(\phi)$.

Q23. A homomorphism $\phi : G \rightarrow H$ is injective if and only if:

- (A) $\text{Im}(\phi) = H$
- (B) $\text{Ker}(\phi) = G$
- (C) $\text{Ker}(\phi) = \{e_G\}$
- (D) $G \cong H$

Answer: (C) $\text{Ker}(\phi) = \{e_G\}$

ϕ is injective $\Leftrightarrow \phi(a) = \phi(b) \Rightarrow a = b \Leftrightarrow \text{Ker}(\phi) = \{e_G\}$ (trivial kernel). Option (A) is surjectivity; (D) requires both injectivity and surjectivity.

Q24. By the First Isomorphism Theorem, if $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ is the natural map $\phi(k) = k \bmod n$, then:

- (A) $\mathbb{Z} \cong \mathbb{Z}_n$
- (B) $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$
- (C) $\text{Ker}(\phi) = \mathbb{Z}_n$
- (D) $\text{Im}(\phi) = \mathbb{Z}$

Answer: (B) $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$

ϕ is surjective with $\text{Ker}(\phi) = n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$. By the First Isomorphism Theorem: $\mathbb{Z}/\text{Ker}(\phi) = \mathbb{Z}/n\mathbb{Z} \cong \text{Im}(\phi) = \mathbb{Z}_n$.

Q25. The group of automorphisms of $\mathbb{Z}_n$, denoted $\text{Aut}(\mathbb{Z}_n)$, is isomorphic to:

- (A) $\mathbb{Z}_n$
- (B) S_n
- (C) $\mathbb{Z}_n^*$ (units mod n)
- (D) $\mathbb{Z}_{n-1}$

Answer: (C) $\mathbb{Z}_n^*$ (the group of units modulo n)

An automorphism of $\mathbb{Z}_n$ is determined by where it sends the generator 1; it must send $1 \mapsto k$ where $\gcd(k, n) = 1$. So $|\text{Aut}(\mathbb{Z}_n)| = \phi(n)$ and $\text{Aut}(\mathbb{Z}_n) \cong (\mathbb{Z}_n^*, \cdot)$.

Q26. Which pair of groups is **isomorphic**?

- (A) $\mathbb{Z}_4$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$
- (B) $\mathbb{Z}_6$ and S_3
- (C) $\mathbb{Z}_4$ and $S_2 \times \mathbb{Z}_2$
- (D) $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4$

Answer: (B) $\mathbb{Z}_6 \cong S_3$? — **Actually NONE of the above isomorphisms hold.**

Careful analysis: $\mathbb{Z}_6$ is Abelian, S_3 is not $\Rightarrow$ **not isomorphic**. $\mathbb{Z}_4$ is cyclic (has element of order 4); $\mathbb{Z}_2 \times \mathbb{Z}_2$ has maximum element order 2 $\Rightarrow$ **not isomorphic**. In a typical exam, $\mathbb{Z}_6 \not\cong S_3$ since one is Abelian and one is not — this tests whether students blindly match cardinalities.

Note: $\mathbb{Z}_4 \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is a **classic exam question**. They have the same order but different structures.

To distinguish groups of the same order: Check (1) Is it cyclic? (2) What are the element orders? (3) Is it Abelian? Two isomorphic groups must have identical element-order structures. $\mathbb{Z}_4$ has an element of order 4; $\mathbb{Z}_2 \times \mathbb{Z}_2$ does not.

Q27. The inner automorphism of G induced by $g \in G$ is the map $\phi_g : G \rightarrow G$ defined by:

- (A) $\phi_g(x) = gx$
- (B) $\phi_g(x) = xg^{-1}$
- (C) $\phi_g(x) = gxg^{-1}$
- (D) $\phi_g(x) = g^{-1}x$

Answer: (C) $\phi_g(x) = gxg^{-1}$

This is called conjugation by g . It is always an automorphism of G . The set of all inner automorphisms $\text{Inn}(G) = \{\phi_g : g \in G\}$ forms a normal subgroup of $\text{Aut}(G)$, and $G/Z(G) \cong \text{Inn}(G)$.

6. Rings, Integral Domains & Fields

A **ring** $(R, +, \cdot)$ is a set with two binary operations satisfying:

- R1.** $(R, +)$ is an Abelian group (identity 0, inverses $-a$)
- R2.** Multiplication is associative: $(ab)c = a(bc)$
- R3.** Distributive laws: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$

Special Rings:

- **Commutative ring:** $ab = ba$ for all a, b .
- **Ring with unity:** has multiplicative identity $1 \neq 0$.
- **Zero divisor:** $a \neq 0$ with $ab = 0$ for some $b \neq 0$.
- **Integral domain:** commutative ring with unity and *no zero divisors*.
- **Field:** commutative ring with unity where every non-zero element has a multiplicative inverse.

Key Formulas & Results

Hierarchy: Field $\subset$ Integral Domain $\subset$ Commutative Ring with Unity $\subset$ Ring

Key Results:

- Every field is an integral domain.
- Every finite integral domain is a field.
- $\mathbb{Z}_p$ is a field $\Leftrightarrow p$ is prime.
- $\mathbb{Z}_n$ is an integral domain $\Leftrightarrow n$ is prime.
- Characteristic of $\mathbb{Z}_n$ is n ; characteristic of $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ is 0.

Integral Domains: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}[x]$, (since $2 \cdot 3 = 0$), $M_n(\mathbb{R})$ ($n \geq 2$), $\mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z}_p$ (prime p)
Not a Field: $\mathbb{Z}$ (2 has no inverse), $\mathbb{Z}_6$
Fields: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ (prime p), $\text{GF}(p^n)$
Rings, NOT Integral Domains: $\mathbb{Z}_6$

6.1. MCQs — Rings, Integral Domains & Fields

Q28. Which of the following is an integral domain but NOT a field?

- (A) $\mathbb{Z}_5$
- (B) $\mathbb{Q}$
- (C) $\mathbb{Z}$
- (D) $\mathbb{R}$

Answer: (C) $\mathbb{Z}$

$\mathbb{Z}$ is an integral domain (commutative, unity, no zero divisors) but it is **not** a field because $2 \in \mathbb{Z}$ has no multiplicative inverse in $\mathbb{Z}$ (since $1/2 \notin \mathbb{Z}$). Options (A), (B), (D) are all fields.

Q29. In the ring $\mathbb{Z}_{12}$, which element is a zero divisor?

- (A) 5
- (B) 7
- (C) 4
- (D) 11

Answer: (C) 4

$4 \cdot 3 = 12 \equiv 0 \pmod{12}$, and both $4 \neq 0$ and $3 \neq 0$ in $\mathbb{Z}_{12}$. So 4 is a zero divisor. Elements 5, 7, 11 all satisfy $\text{gcd}(\cdot, 12) = 1$, so they are **units** (invertible), not zero divisors.

Q30. For which value of n is $\mathbb{Z}_n$ a field?

- (A) $n = 4$
- (B) $n = 6$
- (C) $n = 7$
- (D) $n = 9$

Answer: (C) $n = 7$

$\mathbb{Z}_n$ is a field if and only if n is prime. Among the options: $4 = 2^2$, $6 = 2 \cdot 3$, $9 = 3^2$ are composite; 7 is prime. So $\mathbb{Z}_7$ is a field.

Q31. The characteristic of the field $\mathbb{Q}$ of rational numbers is:

- (A) 1
- (B) 2
- (C) ∞
- (D) 0

Answer: (D) 0

The characteristic of a ring R is the smallest positive n with $n \cdot 1 = 0$; if no such n exists, $\text{char}(R) = 0$. Since $\mathbb{Q}$ has characteristic 0 ($1 + 1 + \cdots \neq 0$ regardless of how many times), the answer is 0.

Q32. Which of the following sets forms a subring of $\mathbb{R}$?

- (A) $\{1, -1\}$
- (B) $\mathbb{Z}$
- (C) $(0, \infty)$ under usual $+$ and $\times$
- (D) $\mathbb{Q}^+$ (positive rationals)

Answer: (B) $\mathbb{Z}$

$\mathbb{Z}$ is closed under addition, subtraction, and multiplication; contains 0 (additive identity) and 1 (multiplicative identity) — all subring requirements are met. Option (A) is not closed under addition ($1 + 1 = 2 \notin \{1, -1\}$). $(0, \infty)$ is not closed under subtraction.

Subring Test: $S \subseteq R$ is a subring iff (1) $S \neq \emptyset$, (2) $a - b \in S$, (3) $ab \in S$ for all $a, b \in S$. Note: a subring need *not* contain the unity of R .

Q33. Every finite integral domain is:

- (A) A ring with zero divisors
- (B) A field
- (C) A non-commutative ring
- (D) Isomorphic to $\mathbb{Z}$

Answer: (B) A field

This is a fundamental theorem: if D is a finite integral domain, then every non-zero element $a \in D$ is a unit. Proof sketch: the map $x \mapsto ax$ is injective (no zero divisors), hence bijective (finite set), so a has an inverse. Thus D is a field.

Q34. The number of elements in the finite field $\text{GF}(p^n)$ is:

- (A) $p \cdot n$
- (B) $p + n$
- (C) p^n
- (D) n^p

Answer: (C) p^n

The Galois field $\text{GF}(p^n)$ (also written $\mathbb{F}_{p^n}$) has exactly p^n elements, where p is prime and $n \geq 1$. Finite fields exist with q elements *if and only if* q is a prime power.

7. Vector Spaces

A **vector space** V over a field F is an Abelian group $(V, +)$ together with scalar multiplication $F \times V \rightarrow V$ satisfying for all $\mathbf{u}, \mathbf{v} \in V$, $\alpha, \beta \in F$:

V1. $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$

V3. $(\alpha\beta)\mathbf{u} = \alpha(\beta\mathbf{u})$

V2. $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}$

V4. $1 \cdot \mathbf{u} = \mathbf{u}$

- $\mathbb{R}^n$ over $\mathbb{R}$: the standard n -dimensional vector space.
- $\mathbb{C}$ over $\mathbb{R}$: a 2-dimensional real vector space.
- $\mathbb{P}_n(\mathbb{R})$: polynomials of degree $\leq n$ over $\mathbb{R}$, dimension $n + 1$.
- $M_{m \times n}(\mathbb{R})$: $m \times n$ matrices over $\mathbb{R}$, dimension mn .
- $C[a, b]$: continuous functions on $[a, b]$ — **infinite dimensional**.
- $\mathbb{R}[x]$: all polynomials over $\mathbb{R}$ — **infinite dimensional**.
- $\mathbb{Z}_p^n$ over $\mathbb{Z}_p$: a finite vector space of dimension n over $\mathbb{Z}_p$, with p^n elements.

7.1. MCQs — Vector Spaces

Q35. Which of the following is **NOT** a vector space over $\mathbb{R}$?

- (A) $\mathbb{R}^3$ with usual operations
- (B) $\{(x, y) \in \mathbb{R}^2 : x + y = 1\}$ with usual operations
- (C) $M_{2 \times 2}(\mathbb{R})$ with matrix addition and scalar mult.
- (D) $\mathbb{P}_2(\mathbb{R})$ (polynomials of degree ≤ 2)

Answer: (B) $\{(x, y) : x + y = 1\}$

This set does not contain the zero vector $(0, 0)$ (since $0 + 0 = 0 \neq 1$) and is not closed under addition ($(1, 0) + (1, 0) = (2, 0)$ but $2 + 0 = 2 \neq 1$). Hence it fails to be a vector space.

Q36. The dimension of the vector space of all 3×3 symmetric matrices over $\mathbb{R}$ is:

- (A) 9
- (B) 3
- (C) 6
- (D) 5

Answer: (C) 6

A 3×3 symmetric matrix $A = (a_{ij})$ with $a_{ij} = a_{ji}$ has 3 diagonal entries (a_{11}, a_{22}, a_{33}) and 3 upper-triangular entries (a_{12}, a_{13}, a_{23}) . Total free entries $= 3 + 3 = 6$.

General formula: $n \times n$ symmetric matrices have dimension $\frac{n(n+1)}{2}$.

Q37. The set $\{1, x, x^2, x^3\}$ is a basis for $\mathbb{P}_3(\mathbb{R})$ because:

- (A) It spans $\mathbb{P}_3$ but is linearly dependent
- (B) It spans $\mathbb{P}_3$ and is linearly independent
- (C) It only spans degree-3 polynomials
- (D) It contains 4 elements equal to $\dim(\mathbb{P}_3) = 3$

Answer: (B) It spans $\mathbb{P}_3$ and is linearly independent

$\dim(\mathbb{P}_3(\mathbb{R})) = 4$ (polynomials of degree ≤ 3). The standard basis $\{1, x, x^2, x^3\}$ has 4 elements, is linearly independent (no non-trivial linear combination is the zero polynomial), and spans all of $\mathbb{P}_3(\mathbb{R})$.

Q38. The number of elements in the vector space $\mathbb{F}_3^2$ (2-dimensional space over $\mathbb{Z}_3$) is:

- (A) 6
- (B) 9
- (C) 3
- (D) 27

Answer: (B) 9

$\mathbb{F}_3^2$ consists of all pairs (a, b) with $a, b \in \mathbb{Z}_3 = \{0, 1, 2\}$. Number of elements $= 3^2 = 9$.

General: An n -dimensional vector space over $\mathbb{F}_q$ (a field with q elements) has q^n elements.

8. Linear Independence, Basis & Dimension

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ are **linearly independent** if:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0} \Rightarrow c_1 = c_2 = \dots = c_k = 0.$$

A **basis** of V is a linearly independent set that **spans** V .

Dimension $\dim(V)$ = number of vectors in any basis (unique).

- Any n linearly independent vectors in an n -dimensional space form a basis.
- Any spanning set of size n in an n -dimensional space is a basis.
- $\dim(\mathbb{R}^n) = n$, $\dim(M_{m \times n}) = mn$, $\dim(\mathbb{P}_n) = n + 1$.
- $\dim(\text{Row space}) = \dim(\text{Column space}) = \text{rank}(A)$.
- $\text{rank}(A) + \text{nullity}(A) = n$ (Rank-Nullity Theorem).

Key Formulas & Results

Rank-Nullity Theorem: For $T : V \rightarrow W$:

$$\dim(\text{Ker } T) + \dim(\text{Im } T) = \dim(V)$$

$$\text{nullity}(T) + \text{rank}(T) = \dim(V)$$

Dimension of subspaces: If U, W are subspaces of V :

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

8.1. MCQs — Linear Independence, Basis & Dimension

Q39. The vectors $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$ in $\mathbb{R}^3$ are:

- (A) Linearly independent and form a basis for $\mathbb{R}^3$
- (B) Linearly dependent
- (C) Linearly independent but do not span $\mathbb{R}^3$
- (D) A basis for $\mathbb{R}^2$

Answer: (C) Linearly independent but do not span $\mathbb{R}^3$

These three vectors all lie in the xy -plane (third coordinate is 0), so they cannot span $\mathbb{R}^3$. However, within the plane they are independent: $c_1(1, 0, 0) + c_2(0, 1, 0) + c_3(1, 1, 0) = \mathbf{0}$ gives $c_1 + c_3 = 0$, $c_2 + c_3 = 0$, and these are independent (check: if $c_3 = t$ then $c_1 = c_2 = -t$, but then $c_1(1, 0, 0) + c_2(0, 1, 0) + c_3(1, 1, 0) = (-t + t, -t + t, 0) = \mathbf{0}$ only if $t = 0$). So they are linearly independent but cannot span $\mathbb{R}^3$.

Q40. If A is a 4×6 matrix of rank 3, then the nullity of A is:

- (A) 1
- (B) 3
- (C) 4
- (D) 2

Answer: (B) 3

By the Rank-Nullity Theorem: $\text{nullity}(A) = n - \text{rank}(A)$ where $n = 6$ is the number of columns.

Nullity = $6 - 3 = 3$.

Q41. Which of the following sets is a basis for $\mathbb{R}^2$?

- (A) $\{(1, 2), (2, 4)\}$
- (B) $\{(1, 0), (0, 1), (1, 1)\}$
- (C) $\{(1, 2), (3, 5)\}$
- (D) $\{(0, 0), (1, 0)\}$

Answer: (C) $\{(1, 2), (3, 5)\}$

(A) The vectors $(1, 2)$ and $(2, 4) = 2(1, 2)$ are linearly dependent (not a basis).

(B) Three vectors in $\mathbb{R}^2$ are always dependent (too many for a 2D basis).

(C) Check: $\det \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix} = 5 - 6 = -1 \neq 0$, so linearly independent and span $\mathbb{R}^2$.

(D) Contains the zero vector, which always makes a set dependent.

Quick Independence Test for $\mathbb{R}^n$: Arrange vectors as rows (or columns) of a matrix and compute the determinant. If $\det \neq 0$, the vectors are linearly independent and form a basis. This is the fastest method on PSC exams.

Q42. The dimension of the null space of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ is:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Answer: (C) 2

Row-reducing A : rows 2 and 3 are linear combinations of row 1 (since $R_2 - 4R_1$ and $R_3 - 7R_1$ give dependent rows). Rank of $A = 2$ (two pivot rows). By Rank-Nullity: nullity $= 3 - 2 = 2$.

(Note: $\det(A) = 1(45 - 48) - 2(36 - 42) + 3(32 - 35) = -3 + 12 - 9 = 0$, confirming rank < 3 .)

Q43. Two subspaces U and W of $\mathbb{R}^4$ have $\dim U = 3$ and $\dim W = 3$. Then $\dim(U \cap W)$ is at least:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Answer: (C) 2

By the dimension formula: $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Since $U + W \subseteq \mathbb{R}^4$: $\dim(U + W) \leq 4$. So $4 \geq 3 + 3 - \dim(U \cap W)$, giving $\dim(U \cap W) \geq 2$.

9. Field Extensions & Finite Fields

If $F \subseteq E$ are fields, then E is an **extension field** of F . E is a vector space over F .

- **Degree of extension:** $[E : F] = \dim_F(E)$ (dimension as F -vector space).
- **Finite extension:** $[E : F] < \infty$.
- $[E : F] = 1 \Leftrightarrow E = F$.
- **Tower Law:** If $F \subseteq K \subseteq E$: $[E : F] = [E : K] \cdot [K : F]$.
- $[\mathbb{C} : \mathbb{R}] = 2$ (basis: $\{1, i\}$); $[\mathbb{R} : \mathbb{Q}] = \infty$.
- Finite fields: $\mathbb{F}_{p^n}$ has degree n over $\mathbb{F}_p$.

Key Formulas & Results

Finite Field Facts:

- A finite field has p^n elements for some prime p and $n \geq 1$.
- Any two finite fields of the same order are isomorphic.
- The multiplicative group $\mathbb{F}_{p^n}^*$ is **cyclic** of order $p^n - 1$.
- $\mathbb{F}_{p^n}$ contains $\mathbb{F}_{p^m}$ as a subfield $\Leftrightarrow m \mid n$.

9.1. MCQs — Field Extensions & Finite Fields

Q44. The degree $[\mathbb{C} : \mathbb{R}]$ of $\mathbb{C}$ as an extension of $\mathbb{R}$ is:

- (A) 1
- (B) 4
- (C) ∞
- (D) 2

Answer: (D) 2

$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$ is a vector space over $\mathbb{R}$ with basis $\{1, i\}$. So $\dim_{\mathbb{R}}(\mathbb{C}) = 2$, i.e., $[\mathbb{C} : \mathbb{R}] = 2$. This is a **degree 2** extension.

Q45. Which of the following is NOT a valid finite field order?

- (A) $4 = 2^2$
- (B) $9 = 3^2$
- (C) $6 = 2 \cdot 3$
- (D) $8 = 2^3$

Answer: (C) 6

Finite fields exist if and only if their order is a **prime power**. $6 = 2 \times 3$ is not a prime power, so no field of order 6 exists. $4 = 2^2$, $9 = 3^2$, $8 = 2^3$ are all prime powers and correspond to valid finite fields.

Q46. If $[E : K] = 4$ and $[K : F] = 3$, then $[E : F]$ equals:

- (A) 7
- (B) 12
- (C) 1
- (D) $\frac{4}{3}$

Answer: (B) 12

By the **Tower Law**: $[E : F] = [E : K] \cdot [K : F] = 4 \times 3 = 12$.

Q47. The multiplicative group of the field $\mathbb{F}_{p^n}^*$ is:

- (A) A cyclic group of order p^n
- (B) A cyclic group of order $p^n - 1$
- (C) An Abelian but non-cyclic group
- (D) A group of order p^{n-1}

Answer: (B) A cyclic group of order $p^n - 1$

$\mathbb{F}_{p^n}^* = \mathbb{F}_{p^n} \setminus \{0\}$ has $p^n - 1$ elements. A fundamental theorem of algebra states this group is **cyclic** — in fact it is generated by a primitive element (primitive root).

Q48. $\mathbb{F}_{p^m}$ is a subfield of $\mathbb{F}_{p^n}$ if and only if:

- (A) $m < n$
- (B) $m = n$
- (C) $m \mid n$
- (D) $\gcd(m, n) = 1$

Answer: (C) $m \mid n$

$\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ iff $m \mid n$. Example: $\mathbb{F}_4 \subset \mathbb{F}_{16}$ since $2 \mid 4$; but $\mathbb{F}_4 \not\subset \mathbb{F}_8$ since $2 \nmid 3$.

10. Mixed & Past Paper Style MCQs

10.1. Comprehensive Mixed Questions

Q49. The number of distinct groups of order 4 up to isomorphism is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Answer: (B) 2

Groups of order 4: (1) $\mathbb{Z}_4$ (cyclic), (2) $\mathbb{Z}_2 \times \mathbb{Z}_2$ (Klein four-group). These are the only two non-isomorphic groups of order 4. $\mathbb{Z}_4$ has an element of order 4; $\mathbb{Z}_2 \times \mathbb{Z}_2$ has all non-identity elements of order 2.

Q50. Every group of order p^2 (for prime p) is:

- (A) Simple
- (B) Non-Abelian
- (C) Abelian
- (D) Isomorphic to $\mathbb{Z}_{p^2}$ only

Answer: (C) Abelian

A classical theorem states: every group of order p^2 (prime p) is Abelian. Moreover, it is isomorphic to either $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$. The proof uses the fact that $Z(G) \neq \{e\}$ for p -groups.

Q51. Which is correct about the alternating group A_n ?

- (A) A_n is the group of all permutations of n elements
- (B) $|A_n| = n!$
- (C) A_n is a normal subgroup of S_n with $|A_n| = \frac{n!}{2}$
- (D) A_n is non-Abelian for all n

Answer: (C) $A_n \trianglelefteq S_n$ with $|A_n| = \frac{n!}{2}$

A_n consists of all even permutations. It has index 2 in S_n , hence is normal, and $|A_n| = n!/2$. Note: $A_1 \cong A_2 \cong \{e\}$ (trivial, Abelian), $A_3 \cong \mathbb{Z}_3$ (Abelian), so option (D) is false.

Q52. In ring theory, a **maximal ideal** M of a commutative ring R with unity satisfies:

- (A) R/M is an integral domain
- (B) R/M is a field
- (C) M is prime but not maximal
- (D) $R/M \cong R$

Answer: (B) R/M is a field

For commutative rings with unity: M is maximal $\Leftrightarrow R/M$ is a field. Compare: M is prime $\Leftrightarrow R/M$ is an integral domain. Every maximal ideal is prime (since every field is an integral domain), but the converse fails in general.

Ideal Correspondence (PSC Favourite):

- M maximal $\Leftrightarrow R/M$ is a **field**
- P prime $\Leftrightarrow R/P$ is an **integral domain**

- Every maximal ideal is prime; converse is false in general.

Q53. The kernel of the ring homomorphism $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ defined by $\phi(k) = k \bmod n$ is:

- (A) $\{0\}$
- (B) $\mathbb{Z}$
- (C) $n\mathbb{Z}$
- (D) $\mathbb{Z}_n$

Answer: (C) $n\mathbb{Z}$

$\text{Ker}(\phi) = \{k \in \mathbb{Z} : \phi(k) = 0\} = \{k \in \mathbb{Z} : n \mid k\} = n\mathbb{Z} = \{\dots, -2n, -n, 0, n, 2n, \dots\}$. By the First Isomorphism Theorem: $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Q54. A linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ can be:

- (A) Injective but not surjective
- (B) Bijective
- (C) Surjective but not injective
- (D) Neither injective nor surjective

Answer: (C) Surjective but not injective

By Rank-Nullity: $\text{rank}(T) + \text{nullity}(T) = 3$. For T to be surjective, $\text{rank}(T) = 2$, forcing $\text{nullity}(T) = 1 > 0$, so T is NOT injective. T cannot be bijective ($\mathbb{R}^3 \not\cong \mathbb{R}^2$). T cannot be injective without $\text{rank} = 3$, but $\text{rank} \leq \dim(\mathbb{R}^2) = 2$.

Q55. The set $W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$ is a subspace of $\mathbb{R}^3$ of dimension:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Answer: (B) 2

W is the null space of the 1×3 matrix $A = [1 \ 2 \ -1]$. $\text{rank}(A) = 1$, so by Rank-Nullity, $\text{nullity} = 3 - 1 = 2$. Basis for W : set $y = 1, z = 0 \Rightarrow x = -2$: $(-2, 1, 0)$; set $y = 0, z = 1 \Rightarrow x = 1$: $(1, 0, 1)$.

Q56. If G and H are groups with $|G| = 15$ and $|H| = 25$, then a homomorphism $\phi : G \rightarrow H$ has image of order:

- (A) 5
- (B) 15

(C) 1 or 5

(D) 25

Answer: (C) 1 or 5

$|\text{Im}(\phi)|$ must divide both $|G| = 15$ and $|H| = 25$ (since $\text{Im}(\phi) \leq H$ and $G/\text{Ker}(\phi) \cong \text{Im}(\phi)$). $\gcd(15, 25) = 5$. So $|\text{Im}(\phi)|$ divides 5, meaning $|\text{Im}(\phi)| \in \{1, 5\}$.

Comprehensive Quick Reference Card

Abstract Algebra & Linear Algebra — Complete Summary

GROUP THEORY

Concept	Condition/Formula	Key Fact
Order of k in $\mathbb{Z}_n$	$\frac{n}{\gcd(n, k)}$	Most tested formula
Generators of $\mathbb{Z}_n$	$\phi(n)$ many	Euler's totient
Lagrange's Theorem	$ H \mid G $	$[G : H] = G / H $
Cyclic group	$G = \langle a \rangle$	Always Abelian
Normal subgroup	$gNg^{-1} = N$	Index 2 $\Rightarrow$ normal
First Isom. Thm	$G/\text{Ker } \phi \cong \text{Im } \phi$	Fundamental theorem

RING & FIELD THEORY

Structure	Extra Conditions	Example
Ring	Assoc. mult., distrib. laws	$\mathbb{Z}, M_n(\mathbb{R})$
Integral Domain	Comm., unity, no zero divs	$\mathbb{Z}, \mathbb{Z}_p$
Field	Int. domain + mult. inverses	$\mathbb{Q}, \mathbb{R}, \mathbb{Z}_p$
$\mathbb{Z}_n$ is a field	iff n is prime	$\mathbb{Z}_7, \mathbb{Z}_{11}$
Finite field order	Must be p^n	$ \mathbb{F}_{p^n} = p^n$
Ideal M maximal	R/M is a field	Key theorem

VECTOR SPACES & LINEAR ALGEBRA

Rank-Nullity: $\text{rank}(T) + \text{nullity}(T) = \dim(V)$

Dimension formula: $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$

Tower Law: $[E : F] = [E : K] \cdot [K : F]$

Finite vector space over $\mathbb{F}_q$ of dim n has q^n elements

Space	Dimension
$\mathbb{R}^n$	n
$M_{m \times n}(\mathbb{R})$	mn
$\mathbb{P}_n(\mathbb{R})$ ($\deg \leq n$)	$n + 1$
Sym. $n \times n$ matrices	$n(n + 1)/2$
Skew-sym. $n \times n$ matrices	$n(n - 1)/2$
$[\mathbb{C} : \mathbb{R}]$	2
$[\mathbb{F}_{p^n} : \mathbb{F}_p]$	n

Best of Luck in Your PSC Examination!

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Group Theory • Ring Theory • Vector Spaces • Linear Algebra • Field Extensions